Linear Algebra — Exercises

Operations modulo n. Groups and fields

Exercise 1. Let n be a fixed positive integer and let a and b be any two integers. Recall that a is congruent to b modulo n if n divides $a - b$. Calculate: a) $17 \bmod 4$ b) $4 \bmod 17$ c) $(-2) \bmod 5$

Exercise 2. For integers x and y we write $x \equiv y \pmod{n}$ (read: x is congruent to y modulo n) if n divides $x - y$. Solve the equations:

a) $2x \equiv 1 \pmod{7}$

b) $2x \equiv 1 \pmod{6}$

c) $5x \equiv 1 \pmod{9}$

d) $x^2 \equiv 3 \pmod{11}$

e) $2x + 3 \equiv 0 \pmod{5}$

f) $x + k \equiv 0 \pmod{n}$

Exercise 3. Prove that for all $n, k, p \in \mathbb{Z}$ we have the following:

a) $(n \bmod k) \bmod k = n \bmod k$

b) $(n + p) \bmod k = \big((n \bmod k) + (p \bmod k)\big) \bmod k$

c) $(np) \bmod k = \big((n \bmod k)\,(p \bmod k)\big) \bmod k$

Exercise 4. Recall that for $a, b \in \mathbb{N}$ the operations of addition and multiplication modulo n are defined by $a +_n b = (a + b) \bmod n$ and $a \cdot_n b = (a \cdot b) \bmod n$. Prove that:

a) addition modulo n is associative, i.e. $(a +_n b) +_n c = a +_n (b +_n c)$ for any $a, b, c \in \mathbb{N}$;

b) multiplication modulo n is associative, i.e. $(a \cdot_n b) \cdot_n c = a \cdot_n (b \cdot_n c)$ for any $a, b, c \in \mathbb{N}$;

c) multiplication modulo n is distributive with respect to addition modulo n, i.e. $(a +_n b) \cdot_n c = (a \cdot_n c) +_n (b \cdot_n c)$ for any $a, b, c \in \mathbb{N}$.

Exercise 5. State which of the following are groups: a) $(\mathbb{Z}_n, +_n)$ b) $(\{1, 2, 3, 4, 6, 12\}, \gcd)$

Exercise 6. State which of the following are groups: a) $(\{z \in \mathbb{C} \mid |z| = 1\}, +)$ b) $(\{z \in \mathbb{C} \mid |z| = 1\}, \cdot)$

Exercise 7. Show that for every n all complex roots of 1 of order n form a group under multiplication. Write down the table of the group of all complex roots of 1 of order 4.

Exercise 8. Show that $(2^X, \triangle)$ is a group, where $\triangle$ denotes the symmetric difference. Construct the group table of $(2^{\{a,b,c\}}, \triangle)$.

Exercise 9. Show that $(\mathbb{Z}_n \setminus \{0\}, \cdot_n)$ is a group if and only if n is a prime number.

Exercise 10. Verify whether $(\mathbb{R}_+, \bullet)$ is a group, where $a \bullet b = a^2 b^2$.

Exercise 11. Let $p \in \mathbb{N}$ be a prime number. Show that $(\mathbb{Z}_p, +_p, \cdot_p)$ is a field.

Exercise 12. Determine which of the following structures are fields. Do this in two steps: first verify whether the set is a group with respect to the first operation, and then whether the set without the identity element of the first operation is a group with respect to the second operation. Here 2^X carries the operations $\cup$, $\cap$ and the symmetric difference $\triangle$, while $\mathbb{R}^{\mathbb{R}}$ denotes the set of all functions $\mathbb{R} \to \mathbb{R}$, with pointwise addition $+$ and composition $\circ$. a) $(2^X, \cup, \cap)$ b) $(\mathbb{R}^{\mathbb{R}}, +, \circ)$ c) $(2^X, \cap, \cup)$

d) $(\mathbb{R}^{\mathbb{R}}, \circ, +)$ e) $(2^X, \cap, \triangle)$ f) $(2^X, \triangle, \cap)$ g) $(2^X, \cup, \triangle)$ h) $(2^X, \triangle, \cup)$

Exercise 13. Verify whether $(\mathbb{R}, \oplus, \odot)$ is a field, where $a \oplus b = a + b + 1$ and $a \odot b = ab + a + b$.